Sequence Model

Types of Sequence Models

- There are two kinds of sequence models:
  1. Scenarios
  2. Sequence diagrams and Collaboration diagrams

- **Scenarios:**

- "A scenario is a sequence of events that occurs during one particular execution of a system"
- Generally written as text statements for a use case

Scenario: Example

- Place Order Use case:
  - Customer logs in
  - System Verifies Customer Credentials
  - Customer searches for specific product
  - System displays products
  - Customer selects specific product
  - System ask for confirmation
  - Customer confirms purchase and and makes payment
  - System verifies payment and confirms order

Scenario

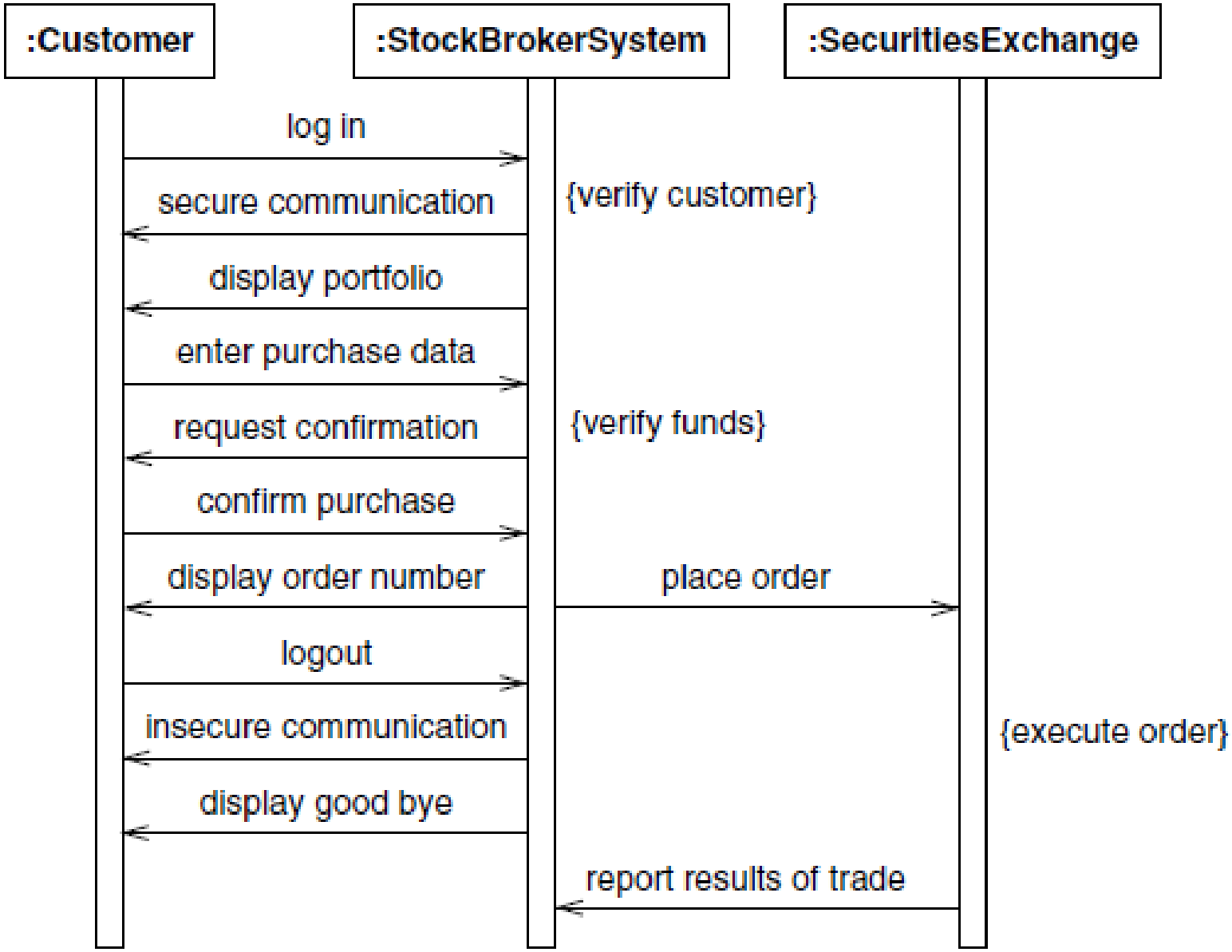
- A scenario contains messages between objects as well as activities performed by objects.
- Each message transmits information from one object to another.
  - For example, Customer logs in transmits a message from Customer to the system.
- The first step of writing a scenario is to identify the objects exchanging messages.
- Then we need to identify messages, as well as the sequence of the messages.

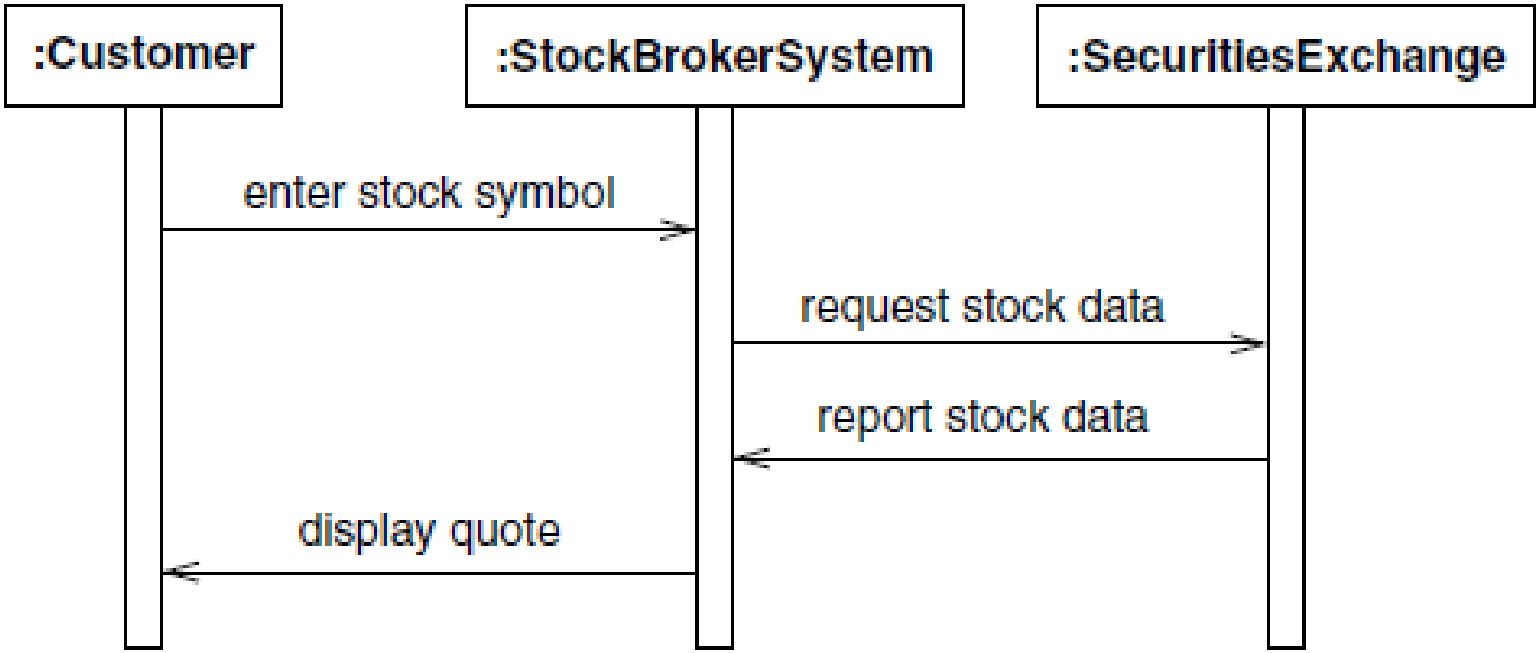
Sequence Diagrams

- “A sequence diagram shows the participants in an interaction and the sequence of messages among them”.
- Generally drawn for a use case using specific notations
- Every use case can have
  - Primary Scenario
  - Alternate Scenarios
- Hence for each use case, there can be multiple sequence diagrams

Key Parts of Sequence Diagrams

- Participant:
  - Object or Entity that acts in a scenario
- Message:
  - Communication between participant objects
  - Can be method calls between two objects
- Axes:
  - Horizontal axis shows which object is interacting
  - Vertical axis shows time


:Customer
:StockBrokerSystem
:SecuritiesExchange
log in
secure communication
{verify customer}
display portfolio
enter purchase data
request confirmation
{verify funds}
confirm purchase
display order number
place order
logout
insecure communication
{execute order}
display good bye
report results of trade


:Customer
:StockBrokerSystem
:SecuritiesExchange
enter stock symbol
request stock data
report stock data
display quote